Zero-Shot Anomaly Detection for Photovoltaic Inverter Fleets via Self-Supervised Contrastive Learning

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Abstract

Scalable zero-shot anomaly detection in photovoltaic inverter fleets is critical for minimizing downtime and ensuring renewable energy stability, yet labeled faults remain scarce and models often fail to generalize across devices. We propose **TRACE**, a self-supervised contrastive learning framework for scalable and label-free fault monitoring across photovoltaic (PV) inverter fleets. TRACE learns transferable embeddings from unlabeled multivariate sensor data by coupling a Transformer autoencoder with a Siamese contrastive network trained using reconstruction-aware positive augmentations and hard negative mining. The pretrained TRACE network generalizes across inverters, deriving latent representations that enable plug-and-play *zero-shot* anomaly detection without retraining. On a real PV dataset (12 inverters, 137 sensors, two weeks of data), TRACE achieves F1=0.993 on unseen inverters versus 0.777 for source-trained baselines, reducing deployment overhead by over 90% and outperforming state-of-the-art time-series AD methods by up to 17%. TRACE demonstrates a scalable path toward AI-enabled, self-supervised monitoring in renewable energy systems.

CCS Concepts

• **Computing methodologies** → **Machine Learning**; **Representation Learning**; **Anomaly Detection**; • **Applied Computing** → **Renewable Energy Systems**.

Keywords

Time series, Self-supervised learning, Contrastive learning, Anomaly detection, Renewable energy

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1 PROBLEM AND MOTIVATION

Photovoltaic inverter fleets generate massive high-dimensional sensor streams, but labeled faults remain scarce. Per-device supervised models and threshold-based detectors require costly retraining and tuning, which hinder large-scale deployment. The inverters account for nearly 40–50% of PV system failures, and their early fault detection remains a bottleneck for grid reliability [6]. Our goal is to learn *universal* representations of normal operation directly from unlabeled data across multiple devices. We propose **TRACE** (**T**ime-series **R**epresentation learning with **A**utoencoder-based **C**ontrastive
Embeddings), a self-supervised contrastive learning framework that learns domain-invariant embeddings transferable to unseen devices for zero-shot anomaly detection (AD) as shown in Figure 1. TRACE eliminates manual labeling and per-site retraining, drastically reducing operational costs.

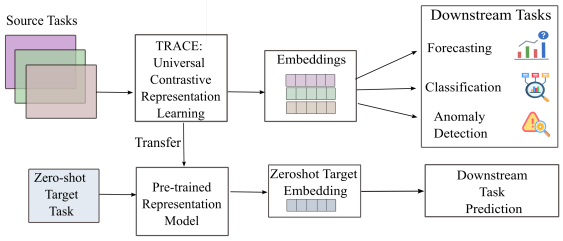

Figure 1: TRACE: universal contrastive framework for renewable energy systems with zero-shot transfer.

2 BACKGROUND AND RELATED WORK

Anomaly detection in multivariate time series. Traditional unsupervised methods such as Isolation Forest [8] and LODA [10] rely on statistical outlier modeling but fail to capture complex temporal dependencies. Deep autoencoder (AE) models and LSTM-based detectors [5] improved feature extraction, yet still depend on reconstruction thresholds that are device-specific. Subsequent generative approaches including DAGMM [16], OmniAnomaly [11], MSCRED [15], and USAD [2] learned richer latent distributions but remained sensitive to data drift. More recent transformer-based models such as Anomaly Transformer [13], InterFusion [7], and TranAD [12] capture long-range dependencies, though their supervised or semi-supervised nature limits cross-device generalization.

Contrastive and self-supervised time series AD. Contrastive learning has recently shown promise for time-series AD by learning robust, label-free representations. TS2Vec [14] introduces hierarchical temporal contrast, while CARLA [4] employs adversarial negative samples to improve discriminability. GANF [3] and RAN-SynCoders [1] integrate generative or graph priors to synthesize hard negatives. However, most rely on handcrafted traditional time series augmentations—jittering, warping, or scaling—that distort temporal coherence and hinder transferability.

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Renewable energy applications. In PV systems, previous studies have primarily used LSTM or AE architectures for inverter fault detection [9, 7], but each model must be retrained for new devices or operating conditions. TRACE differs by learning reconstruction-driven, semantically consistent augmentations that generalize across inverters without labels, enabling zero-shot deployment in renewable energy environments.

3 APPROACH AND UNIQUENESS

TRACE integrates two core sequential modules as shown in Figure 2:

(1) **Autoencoder Pretraining.** A Transformer-based autoencoder learns a latent representation $\mathbf{z} = f_\theta(\mathbf{x})$ of windowed sensor data $\mathbf{x} \in \mathbb{R}^{T\times d}$, that filters noise, seasonality, and site-specific variations by minimizing:

$$L_{rec} = \frac{1}{T}\sum_{t=1}^{T} \|\mathbf{x}_t - \hat{\mathbf{x}}_t\|_2^2.$$

(2) **Contrastive Siamese Training.** Using reconstructions $\hat{\mathbf{x}}$, we generate positive augmentations $\mathbf{x}^+ = \hat{\mathbf{x}} + \epsilon$ ($\epsilon \sim \mathcal{N}(0, \sigma^2 I)$) and hard negatives from random, distance, or reconstruction-error criteria. Pairs $(\mathbf{x}, \mathbf{x}^\pm)$ are passed through shared encoder $\mathcal{F}_\theta$, and embeddings $\mathbf{h}, \mathbf{h}^\pm$ are optimized under contrastive loss:

$$L_{contrast} = \sum_i \left[y_i \tfrac{1}{2} d_i^2 + (1-y_i)\tfrac{1}{2}\max(0, m - d_i)^2 \right],$$

where $d_i = \|\mathbf{h} - \mathbf{h}^\pm\|_1$.

For AD, the embeddings from inverters are clustered using K-Means or DBSCAN. The dense clusters correspond to normal behavior, while the sparse ones correspond to anomalies. Algorithm 1 presents the two-stage training procedure of TRACE, and the open-source implementation is provided for reproducibility.[1]

4 RESULTS AND CONTRIBUTIONS

4.1 Experimental setup

TRACE is evaluated on a fleet of 12 grid-connected PV inverters deployed in Germany (137 sensors, two weeks of training data). Ten inverters serve as source devices; two are reserved for zero-shot target testing. Sliding windows ($T = 10$) are normalized per channel. The AE uses 64-dimensional latent space and both modules are trained for 200 epochs with batch size 128. Clustering is performed using K-Means (Silhouette-based k) and DBSCAN (validated ϵ, minPts).

4.2 Comparison with baselines

We compare TRACE with a diverse range of state-of-the-art time-series AD methods, spanning traditional statistical, recurrent, generative, and transformer-based architectures. As shown in Table 1, TRACE achieves the highest overall average score (0.90) and ranks first across all metrics. The closest competitor, GANF [3], attains an average score of 0.88, followed by Anomaly Transformer [13] and TranAD [12] at 0.86–0.87. Compared to the strongest baseline, TRACE yields a relative improvement of 2.3% and surpasses the mean of all baselines by approximately 17%. Traditional unsupervised models such as Isolation Forest [8] and LSTM [5] underperform under domain shifts, while OmniAnomaly [11] and Anomaly Transformer [13] effectively capture temporal dependencies but lack transferability. Recent contrastive framework CARLA shows progress but remains sensitive to augmentation choices. TRACE consistently outperforms all these methods, benefiting from its AE-driven positive sampling and reconstruction-error-based hard negatives, which preserve temporal coherence and enhance anomaly separability in the learned embedding space.

[1] https://github.com/anonymousgit234/TRACE-Time-Series-Representation-Learning-with-Contrastive-Embeddings-for-Anomaly-Detection

Algorithm 1 TRACE: Two-stage training and evaluation

Require: Training data $\mathcal{D}_{train}$, test data $\mathcal{D}_{test}$, epochs E_{AE}, E_{CL}, batch size B, margin m

```
Stage 1: Autoencoder pretraining
for epoch = 1 to E_AE do
    for batch {x_a} from D_train do
        Encode z_a = f_θ(x_a); reconstruct x̂_a = g_φ(z_a)
        Update (θ, φ) to minimize L_rec
    end for
end for
Stage 2: Contrastive Siamese training
for each anchor x_a ∈ D_train do
    Generate positives x_a^+ and negatives x_a^-
end for
for epoch = 1 to E_CL do
    for batch of pairs (x_a, x_a^±, y_i) do
        Compute embeddings h_a, h_a^± = F_θ(·)
        Update parameters using L_contrast
    end for
end for
Stage 3: Evaluation: Cluster embeddings for anomaly labels.
```

Table 1: Comparison with multivariate time series AD benchmark methods.

Model	F1	Precision	Recall	Avg	Avg Rank
Isolation Forest [8]	0.76	0.86	0.71	0.77	9
LSTM [5]	0.82	0.85	0.83	0.83	6.5
DAGMM [16]	0.70	0.92	0.64	0.75	10
MSCRED [15]	0.44	0.98	0.35	0.59	12
OmniAnomaly [11]	0.81	0.82	0.85	0.83	6.5
AnomalyTransformer [13]	0.85	0.89	0.87	0.87	3
Interfusion [7]	0.30	**1.00**	0.19	0.49	14
TranAD [12]	0.85	0.89	0.84	0.86	4.5
USAD [2]	0.85	0.90	0.83	0.86	4.5
RANSynCoders [1]	0.58	0.88	0.51	0.65	11
GANF [3]	0.86	0.89	**0.90**	0.88	2
LODA [10]	0.80	0.91	0.75	0.82	8
CARLA [4]	0.45	0.31	**0.90**	0.55	13
TRACE (Ours)	**0.87**	0.96	0.88	**0.90**	**1**

Table 2: Zero-shot anomaly detection performance on source vs target inverters.

Method	Inverter Set	Precision	Recall	F1
K-Means	Source	0.738	0.930	0.777
K-Means	Target	**0.994**	**0.992**	**0.993**
DBSCAN	Source	0.624	0.931	0.690
DBSCAN	Target	**0.842**	**0.998**	**0.909**

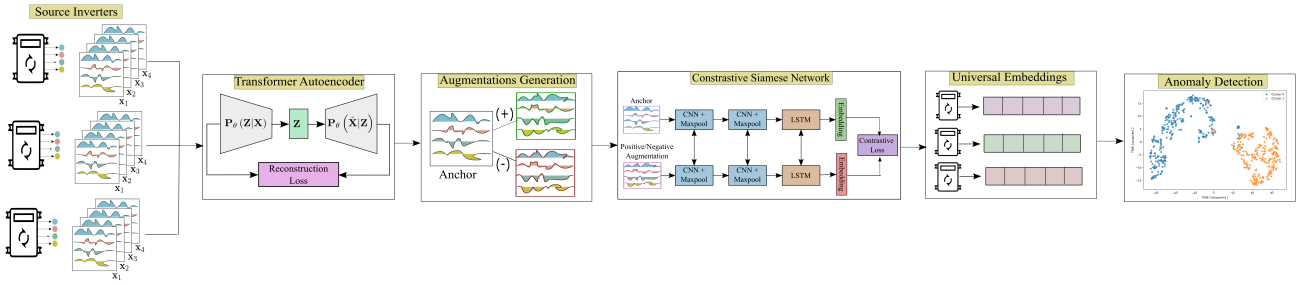

Figure 2: Architecture of TRACE: AE pretraining, reconstruction-based augmentations, contrastive Siamese embedding, and clustering for anomaly detection.

4.3 Zero-shot transfer performance

We next evaluate the ability of TRACE to generalize to unseen PV inverters without retraining. The model is trained on ten source inverters and tested in a zero-shot setting on two previously unseen target inverters. As shown in Table 2, TRACE achieves an F1 score of 0.993 on the target set using K-Means clustering, compared to 0.777 for source-trained baselines. DBSCAN yields consistent results with F1=0.909, confirming the robustness of the learned embeddings. These results demonstrate that TRACE captures device-agnostic temporal signatures that remain invariant to differences in site conditions, irradiance patterns, and inverter characteristics. These results demonstrate that TRACE captures device-agnostic temporal signatures and enables plug-and-play fault monitoring across inverters, reducing deployment overhead by replacing N per-device models with a single universal model, achieving more than 90% savings in training and maintenance effort.

5 CONCLUSION AND FUTURE WORK

TRACE establishes a scalable paradigm for label-free condition monitoring across renewable energy fleets. By learning domain-invariant temporal embeddings, it eliminates per-device retraining and achieves reliable cross-site AD, directly supporting the operational sustainability of PV systems. The same framework can extend to wind turbines, storage units, and hybrid microgrids, enabling unified, AI-driven renewable monitoring.

Future work will focus on integrating graph-based relational priors between assets, incorporating physics-informed constraints into the contrastive objective, and developing explainable embedding visualizations for operator trust. By replacing N per-device models with a single universal network, TRACE moves toward resilient, intelligent energy infrastructures powered by self-supervised learning.

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